Small but Mighty: Test Suites as Improvement Signals for Self-Evolving Agents

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HarnessRSI 64 ★★

evidence: retained

Paper

Code

1 Evolve the gate, not the weights

1 Seed suites → size niches
Frozen LLM writes pytest from module source.
Tests include fixtures, stubs and exact assertions.
Build grid after seeds; re-file by Pareto quality.

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2 Aim at a size; choose parents
UCB allocates calls; kernel corrects count bias.
Nearest front suite on each side of target;
show stronger parent last, then compress/expand.

▼

3 Run coverage + mutation checks
Feedback: uncovered lines and surviving mutants.
Quality = coverage × mutation-kill score (hv).
Score passing tests; size counts written tests.

▼

4 Retain a front at each actual size
Keep nondominated coverage/mutation trade-offs.
Off-target suites enter their landed-size niche.
Failed measurements still consume call budget.

▼

5 Deliver one regression signal
Choose highest hv; smaller size breaks ties.
Then test later rewrites of the original module.
Reject if previously passing tests now fail.

Two keep rules: search retains suite fronts; the later gate rejects regressing harness edits. Passing the gate does not prove an improvement in task success.

2 Higher suite quality, fewer tests

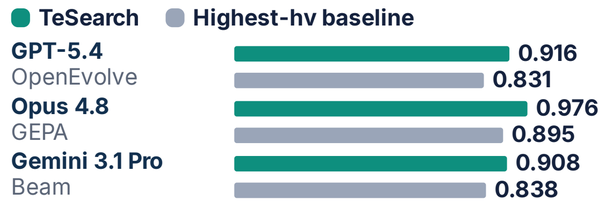


Table 1 · mean hv over 129 modules; 50 calls/module. Held-out mutants, same modules. No SD/CI or repeat count per cell; full suites are smallest among LLM methods.

4 Stronger signal ≠ complete safeguard

Revision gate · Table 2, mini-swe-agent

Pooled	Repo tests	TeSearch
Divergent rejected ↑	25.3%	**59.9%**
False alarms ↓	0.000	**4.0%**
Mean seconds ↓	5.9	3.5

- GPT-5.4 writes revisions; 470 human-verified edits from 512 proposals. Suite-generation backbone for this gate is not specified.
- Uneven gains: content_string 0.057 → 0.600; agent loop 0.529 → 0.588. Cache detection still 0.414; model client 0.421.
- Speed not universal: tool-call parser 3.0 → 3.1 s; config merger 2.9 → 3.1 s. No CIs or longitudinal task-success result.

3 Controls and missing checks

Ablation · Opus 4.8 (Table 1)

Variant	hv ↑	Tests ↓
Full	**0.976**	**29.1**
No kernel	0.917	40.8
No UCB	0.960	34.4
Random parents	0.934	33.2
No size pressure	0.954	**84.6**

- **Not just one model:** every ablation lowers mean hv and enlarges mean suites on all backbones; no variance or per-module breakdown.
- **Access:** OpenEvolve gets same feedback; Claude Code / Cursor get harness tool feedback. Other baseline wiring is not detailed; all LLM methods share 50 calls.
- **Exceptions:** Cursor on Gemini is already small (37.0 tests). Pynguin: 20.4 tests, hv 0.090; only 40 supported modules, 6000 s budget.
- **Split:** search/eval mutants differ, module source is shared. Tests infer expected values from source; no revision-pool contamination audit reported.
- **Tuning:** kernel spread uses an unidentified holdout. Other selection splits and baseline tuning budgets are not reported.
- **Cost:** 3.5 s is gate execution, not suite construction. No model call at gate time; generation dollars/tokens/time not itemized.

Takeaway

Search for a compact regression signal before trusting self-edits. TeSearch catches more behavior changes, but also rejects equivalent edits. A stronger gate is not yet evidence of recursive performance gains.

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Reports: held-out mutants, call matching, ablations, gate latency.
Not reported: search-seed uncertainty, transfer, leakage audit, full generation cost, long-run agent gains.